ENTERPRISE QA & VERIFICATION MANUAL

NTS LEGAL PRO

End-to-End Application Testing Guide & QA Verification Manual

Target System:	**NTS Legal Pro Multi-Tenant Legal Platform**	**Document Version:**	**v2.4.0-Enterprise (Build 2026.10)**
Client Architecture:	Flutter (iOS, Android, Web, Desktop) + Riverpod + GoRouter	**Classification:**	**CONFIDENTIAL / INTERNAL QA**
Live Backend API:	`https://api.hphighcourt.site/api/`	**Target Commit:**	`fa64f951c924 (v2.4.0)`
Testing Scope:	**21 Functional Modules | 5 User Roles | 65+ Formal Test Cases**	**Sign-off Date:**	October 2026

EXECUTIVE PURPOSE & SCOPE OF THIS TESTING SUITE:
This manual is the definitive, multi-role testing and quality assurance specification for the **NTS Legal Pro** Flutter application. It delivers a comprehensive methodology to validate all features across all five system personas: **Platform Super Administrator**, **Firm Managing Administrator**, **Advocate / Lawyer**, **Legal Assistant / Staff**, and **End-Client (Client Portal)**.

Every test case provides precise operational steps, exact input parameters, live backend API interactions, expected interface state changes, and explicit regression verification criteria covering recent critical updates: **Client Document Portal**, **Cryptographic SHA-256 Document Security**, **WhatsApp Dynamic Dispatch**, **Reactive Firm Paywalls (FIRM_BLOCKED / SUBSCRIPTION_EXPIRED)**, **Razorpay Platform Billing**, and **Strict Multi-Tenant Isolation & Role-Based Access Control (RBAC)**.

DOCUMENT OUTLINE & ROADMAP

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QA Lead Reviewer: _____

Lead Engineer: _____

Test Execution Date: ____/____/2026

SECTION 1: TESTING ARCHITECTURE, ENVIRONMENTS & CREDENTIALS

1.1 Architecture & Core Foundation

The NTS Legal Pro client application is built upon Flutter with strict architectural decoupling:
• **Multi-Tenant Isolation:** All firm queries, mutations, document repositories, and communications are partitioned by firm tenant ID (`firm_id`). Under no circumstances may data leak across tenant boundaries.
• **Role-Based Access Control (RBAC):** Dynamic shell navigation, screen gates (`RoleGate`, `CapabilityGate`), and route guards enforce authorized access prior to any screen load or API call.
• **Reactive Paywall Interceptors:** Global HTTP error interceptors watch for 403 response payloads carrying machine codes FIRM_BLOCKED and SUBSCRIPTION_EXPIRED, instantly raising full-screen blocking modal barriers that halt operations until administrative clearance or plan renewal.
• **Observability & Tracing:** Every HTTP request to https://api.hphighcourt.site/api/ captures the X-Request-ID header. Correlation chips in the UI permit one-tap copying for triage.

1.2 Verified Live Test Credentials & Personas

The following credentials have been validated against the live production backend for QA execution:

System Persona	Username / Email	Password	Firm ID	Testing Role & Scope of Authority
Firm Admin	`testlawyer_fe69b5`	`TestPassword@123`	15	Managing Partner. Full financial, litigation, team, subscription, and branding authority.
Lawyer / Advocate	`lawyer_7cc7d`	`TestPassword@123`	15	Practicing Legal Counsel. Cases, documents, hearings, clients. Barred from financials & team billing.
Legal Staff	`staff_7cc7d`	`TestPassword@123`	15	Paralegal / Coordination. Primary Tasks tab, hearings, client intake. Barred from Cases & Documents (403).
Client Portal	`Dynamic Invite Token / client_user`	`User Defined / TestPassword@123`	15	End-Client Portal. Dedicated shell (`/client/dashboard`), hearings countdown, assigned counsel, secure 300s token viewer.
Super Admin	`superadmin`	`PlatformAdmin@2026`	System	Platform Owner. Platform plans, tenant firms registry, firm block/unblock, Razorpay gateway config, audit logs.

1.3 Core Network & Security Endpoints

Category	HTTP Route	Auth Requirement	Expected Behavior / Envelope
Standard Auth	`POST /api/auth/login/`	Public	Returns JWT access, refresh tokens, user profile, role, and firm ID
Client Auth	`POST /api/client-portal/login/`	Public	Dedicated portal login; sets client session and routes to `/client/dashboard`
Client Invite	`POST /api/clients/:id/invite/`	Firm Admin / Lawyer	Generates tokenized URL (validating zero localhost host leakage)
Signed Doc Token	`GET /api/client-portal/documents/:id/access-link/`	Client Bearer	Returns single-use token expiring in precisely 300 seconds (5 minutes)
Doc Streaming	`GET /api/client-portal/documents/download/:id/?token=...`	Query Token	Streams binary PDF bytes directly into in-app viewer or device storage
WhatsApp Send	`POST /api/communications/whatsapp/send/`	Firm Roles	Dispatches Meta Cloud API payload with Click-to-Chat fallback
Paywall Intercept	`GET /api/cases/ (Blocked Firm)`	Any Token	403 Forbidden with errorCode: 'FIRM_BLOCKED' triggers blocking overlay

SECTION 2: ROLE-BASED ACCESS CONTROL (RBAC) & CAPABILITY MATRIX

The following matrix outlines all 18 functional capability areas implemented across the Flutter application and enforces strict validation between the 5 user roles. In addition to hiding UI controls, all blocked routes are guarded by RoleGuard in app_router.dart and reject access with immediate redirection or 403 Forbidden error handling.

Functional Capability Area	Super Admin	Firm Admin	Lawyer	Staff	Client	Live Backend & Router Guard Contract
Platform Tenant Registry	**FULL**	BLOCKED	BLOCKED	BLOCKED	BLOCKED	/app/firms: Redirects non-superadmins to dashboard
Payment Gateway Config	**FULL**	BLOCKED	BLOCKED	BLOCKED	BLOCKED	/admin/payment-gateway/: Super Admin Razorpay keys
Subscription Audit Log	**FULL**	BLOCKED	BLOCKED	BLOCKED	BLOCKED	/admin/subscription-transactions/: Platform audit
Executive Financials	BLOCKED	**FULL**	403 BLOCKED	403 BLOCKED	BLOCKED	GET /api/payments/: Returns 403 for non-admin
Record & Collect Payments	BLOCKED	**FULL**	403 BLOCKED	403 BLOCKED	BLOCKED	POST /api/payments/: Invoicing and billing records
Subscription & Quotas	PLANS ONLY	**FULL**	BLOCKED	BLOCKED	BLOCKED	/app/more/subscriptions: Meter and Razorpay plan renewal
Team & User Management	PROVISION	**FULL**	BLOCKED	BLOCKED	BLOCKED	POST /api/auth/add-user/: Invite advocate & staff
Firm Website & Branding	BLOCKED	**FULL**	BLOCKED	BLOCKED	BLOCKED	/api/profile/firm/website/: Tagline, bio, theme, colors
Cases Litigation Hub	BLOCKED	**FULL**	**FULL**	403 BLOCKED	MY CASES	GET /api/cases/: 403 Forbidden for staff; clients see own
Confidential Case Files	BLOCKED	**FULL**	**FULL**	403 BLOCKED	SHARED ONLY	GET /api/documents/: Staff barred; client sees shared
Document Sharing & Revoke	BLOCKED	**FULL**	**FULL**	BLOCKED	BLOCKED	/clients/:id/documents/:docId/share/: Time-bound share
Client Registry & Intake	BLOCKED	**FULL**	**FULL**	**FULL**	BLOCKED	GET/POST /api/clients/: All firm members can manage clients
Court Hearings & Docket	BLOCKED	**FULL**	**FULL**	**FULL**	MY HEARINGS	GET /api/hearings/: Full schedule for firm; client sees own
Operational Tasks Board	BLOCKED	**FULL**	**ASSIGNED**	**PRIMARY TAB**	BLOCKED	/app/tasks: Primary destination in shell for staff role
WhatsApp Dispatch	BLOCKED	**FULL**	**FULL**	**FULL**	BLOCKED	Template selection, param interpolation, Meta Cloud & Click-to-Chat
Internal Colleague Chat	BLOCKED	**FULL**	**FULL**	**FULL**	BLOCKED	/app/more/messages: Internal messaging with case attaches
Client Portal Experience	BLOCKED	BLOCKED	BLOCKED	BLOCKED	**DEDICATED**	/client/*: Isolated navigation shell, countdown, 300s token
User Profile & Settings	PLATFORM	FIRM	LAWYER	STAFF	CLIENT	/api/profile/me/: Personal settings & theme preferences

SECTION 3: GRANULAR STEP-BY-STEP TEST SUITES

Suite TC-01: Authentication, Onboarding & User Lifecycle

TC-01.1: Standard Multi-Role Sign-In & JWT Session Storage

Target Persona: Firm Admin, Lawyer, Staff | **Priority:** P0 - Critical

Objective: Verify credentials authentication against live backend and correct JWT persistence.

Step-by-Step Execution:
1. Open application to `/login`.
2. Input username `testlawyer_fe69b5` and password `TestPassword@123`.
3. Tap 'Sign In' button.
4. Repeat test for `lawyer_7cc7d` and `staff_7cc7d`.

Expected Application State:
• Network sends `POST /api/auth/login/` returning 200 OK with `access` & `refresh` tokens.
• Secure storage securely caches JWT tokens.
• App transitions immediately to `/app/dashboard`.

Verification Checkpoints:
[ ] Tokens stored securely | [ ] Header role badge matches persona | [ ] No UI latency

TC-01.2: Invalid Credentials & Form Validation

Target Persona: All Roles | **Priority:** P1 - High

Objective: Verify system gracefully handles invalid passwords without crashing or uncaught exceptions.

Step-by-Step Execution:
1. At `/login`, enter username `testlawyer_fe69b5` and incorrect password `WrongPass!99`.
2. Tap 'Sign In'.
3. Clear fields and tap 'Sign In' with empty values.

Expected Application State:
• Client displays red inline error: 'Invalid username or password'.
• Form highlights required fields with validation warnings.
• Application remains stable without unhandled exception overlays.

Verification Checkpoints:
[ ] Clear inline feedback | [ ] Form disabled during in-flight request | [ ] Password obscured

TC-01.3: New Firm Registration & Onboarding

Target Persona: Firm Admin | **Priority:** P1 - High

Objective: Verify law firm signup workflow registers firm entity and initializes administrator.

Step-by-Step Execution:
1. Tap 'Register New Firm' on login view.
2. Enter Firm Name: 'Alpha Legal LLP', Admin Username: 'alphalead_01', Email, and Password.
3. Tap 'Complete Registration'.

Expected Application State:
• Endpoint `POST /api/auth/register/` creates firm record.
• Returns 201 Created and directs user to onboarding dashboard with trial subscription.

Verification Checkpoints:
[ ] Firm created in DB | [ ] Initial quota initialized | [ ] Welcome guide triggered

TC-01.4: Firm Member Invitation Acceptance	**Target Persona:** Invited Lawyer / Staff | **Priority:** P1 - High
Objective: Verify invited team members can activate their account via query token link.	
Step-by-Step Execution: 1. Open link `/accept-invite?token=VALID_FIRM_INVITE_TOKEN`. 2. Verify email is locked to invited address. 3. Enter full name and set secure password; tap 'Activate Account'.	
Expected Application State: • Endpoint `POST /api/auth/accept-invite/` validates token and provisions user. • Logs user in automatically and opens firm dashboard with assigned role permissions.	
Verification Checkpoints: [ ] Pre-filled email correct | [ ] Password strength enforced | [ ] Redirects to dashboard	

Suite TC-02: Role-Adaptive Command Centers & Dashboards

TC-02.1: Firm Admin Executive Command Center	**Target Persona:** Firm Admin (`testlawyer_fe69b5`) | **Priority:** P0 - Critical
Objective: Verify executive dashboard delivers financial metrics, litigation summaries, and quick actions.	
Step-by-Step Execution: 1. Sign in as Firm Admin. 2. Inspect the `/app/dashboard` surface.	
Expected Application State: • Metrics display Active Cases, Total Clients, Pending Hearings, Revenue Collected (■). • Quick Action buttons present: 'New Case', 'New Client', 'Schedule Hearing', 'Record Payment'. • Financial charts and recent transactions rendered cleanly.	
Verification Checkpoints: [ ] Revenue amounts formatted in INR | [ ] Quick actions navigate correctly | [ ] Watermark visible	

TC-02.2: Lawyer Litigation Workspace Dashboard	**Target Persona:** Lawyer (`lawyer_7cc7d`) | **Priority:** P0 - Critical
Objective: Verify advocate dashboard provides litigation-focused agenda and hides financial metrics.	
Step-by-Step Execution: 1. Sign in as Lawyer. 2. Inspect the `/app/dashboard` surface.	
Expected Application State: • Displays Courtroom Agenda (Upcoming hearings this week with court number & judge). • Displays Active Briefs filed and assigned legal tasks. • Financial revenue metrics and 'Record Payment' buttons are completely omitted.	
Verification Checkpoints: [ ] No revenue cards present | [ ] Quick actions omit billing | [ ] Court agenda accurate	

TC-02.3: Staff Operations Command Center Dashboard	**Target Persona:** Staff (`staff_7cc7d`) | **Priority:** P0 - Critical
Objective: Verify paralegal dashboard highlights operational tasks queue and hearing coordination.	
Step-by-Step Execution: 1. Sign in as Staff. 2. Inspect `/app/dashboard`.	
Expected Application State: • Highlights Pending Tasks Queue with interactive status toggles. • Today's Hearings card visible for client notice delivery and courtroom coordination. • Navigation bottom rail replaces 'Cases' with 'Tasks Queue' as direct primary tab.	
Verification Checkpoints: [ ] Primary Tasks destination present | [ ] Cases tab absent | [ ] One-tap task checkbox operational	

Suite TC-03: Legal Cases Management & Proceedings

TC-03.1: Case Directory Listing, Filtering & Search	**Target Persona:** Firm Admin, Lawyer | **Priority:** P0 - Critical
Objective: Verify case management repository supports multi-parameter filtering and search.	
Step-by-Step Execution: 1. Navigate to `/app/cases`. 2. In search bar, type existing case title or CNR number. 3. Filter by Status ('In Progress') and Court ('High Court').	
Expected Application State: • Case list queries `GET /api/cases/` with query params. • Filtered list updates in real-time matching criteria. • Status chips display appropriate color codes (Green for Decided, Blue for In Progress).	
Verification Checkpoints: [ ] Dynamic search response < 300ms | [ ] Filter chips dismissible | [ ] Pagination functioning	

TC-03.2: Case Creation Form & Client Association	**Target Persona:** Firm Admin, Lawyer | **Priority:** P0 - Critical
Objective: Verify complete case initiation workflow associating client, court, and advocates.	
Step-by-Step Execution: 1. From `/app/cases`, tap '+' Floating Action Button. 2. Select Client from dropdown or autocomplete search. 3. Fill Case Title: 'State vs Kumar', Case Number: 'WP/2026/894', Court: 'High Court', Judge: 'Hon. Justice Verma'. 4. Tap 'Save Case'.	
Expected Application State: • Payload sent via `POST /api/cases/`. • Response 201 Created returns newly generated case record. • App navigates to `/app/cases/:id` detail screen.	
Verification Checkpoints: [ ] Client linked properly | [ ] Validation blocks empty title | [ ] Detail view populated	

TC-03.3: Staff RBAC Route Barrier on Cases (403 Verification)	**Target Persona:** Staff (`staff_7cc7d`) | **Priority:** P0 - Security Critical
Objective: Verify Staff persona is strictly barred from accessing cases via UI or direct URL.	
Step-by-Step Execution: 1. Log in as Staff. 2. Attempt to navigate directly to `/app/cases` or `/app/cases/1` via deep link or URL. 3. Observe router guard response.	
Expected Application State: • `RoleGuard.evaluate()` intercepts navigation before screen renders. • Immediately redirects user back to `/app/dashboard`. • Direct API `GET /api/cases/` returns HTTP 403 Forbidden.	
Verification Checkpoints: [ ] Zero confidential case leakage | [ ] Smooth redirect without crash | [ ] 403 captured cleanly	

Suite TC-04: Client Registry, Intake & Portal Invitations

TC-04.1: Client Intake Form & Directory Search	**Target Persona:** Firm Admin, Lawyer, Staff | **Priority:** P0 - Critical
Objective: Verify client directory listing and new client registration workflow.	
Step-by-Step Execution: 1. Navigate to `/app/clients`. 2. Tap 'Add Client'. 3. Enter Name: 'Rajesh Sharma', Email: 'rajesh.sharma@example.com', Phone: '+919876543210', Address. 4. Tap 'Save Client'.	
Expected Application State: • Payload sent via `POST /api/clients/`. • Client appears in directory with status 'Active'. • Search bar immediately locates 'Rajesh Sharma'.	
Verification Checkpoints: [ ] Phone number formatted | [ ] Duplicate email warning if existing | [ ] Detail screen links	

TC-04.2: Client Portal Invitation Flow (No Localhost Link Leak)	**Target Persona:** Firm Admin, Lawyer | **Priority:** P0 - Regression Critical
Objective: Verify invitation dialog generates valid production token URL without localhost leakage.	
Step-by-Step Execution: 1. Open client detail page `/app/clients/:id`. 2. Tap 'Invite to Portal' action button. 3. In dialog, confirm client email and tap 'Generate Client Invite'. 4. Inspect the generated invitation link.	
Expected Application State: • Backend triggers `POST /api/clients/:id/invite/`. • Generated link MUST match https://.../client/accept-invite?token=... or deep link scheme. • Link must NEVER contain http://localhost or 127.0.0.1. • Tap 'Copy to Clipboard'; verify snackbar confirms copy.	
Verification Checkpoints: [ ] Token length > 24 chars | [ ] Zero localhost strings | [ ] Copy to clipboard operational	

TC-04.3: Client Communication Preferences & WhatsApp Opt-In

Target Persona: Firm Admin, Lawyer | **Priority:** P1 - High

Objective: Verify client notification channel preferences modal and WhatsApp consent storage.

Step-by-Step Execution:
1. On client detail view, tap 'Notification Preferences' / WhatsApp icon.
2. Toggle 'Explicit WhatsApp Consent' to TRUE.
3. Set Preferred Channel to 'Both' (WhatsApp & Email).
4. Enter dedicated WhatsApp mobile: '+919876543210' and save.

Expected Application State:
• Endpoint `PUT /api/clients/:id/communication-preferences/` records consent.
• WhatsApp composer icon enables across case hearings and document modals.

Verification Checkpoints:
[ ] Switch states persist on reload | [ ] Audit timestamp recorded | [ ] Number validated

Suite TC-05: Court Hearings & Docket Scheduling

TC-05.1: Schedule Court Hearing with Dynamic Countdown Chip

Target Persona: Firm Admin, Lawyer, Staff | **Priority:** P0 - Critical

Objective: Verify scheduling court hearings, judge allocation, and dynamic countdown computation.

Step-by-Step Execution:
1. Navigate to `/app/hearings`.
2. Tap '+' button.
3. Link Case 'State vs Kumar', Hearing Date (set tomorrow's date), Court 'Room 4', Judge 'Hon. Sharma'.
4. Set Stage: 'Final Arguments' and save.

Expected Application State:
• Endpoint `POST /api/hearings/` registers hearing.
• Hearing tile renders dynamic countdown chip: **TOMORROW** (or **TODAY** / **IN X DAYS**).
• Tile appears in upcoming hearings list ordered chronologically.

Verification Checkpoints:
[ ] Countdown chip updates accurately | [ ] Case details link works | [ ] Staff can schedule

TC-05.2: One-Tap WhatsApp Hearing Reminder Dispatch

Target Persona: Firm Admin, Lawyer, Staff | **Priority:** P1 - High

Objective: Verify hearing detail screen allows sending template-based WhatsApp reminders.

Step-by-Step Execution:
1. Open hearing detail screen `/app/hearings/:id`.
2. Tap 'Send WhatsApp Reminder' button.
3. In composer modal, verify pre-filled parameters: Client Name, Hearing Date, Court Room.
4. Tap 'Send Reminder via WhatsApp'.

Expected Application State:
• Endpoint `POST /api/communications/whatsapp/send/` executes.
• Message dispatched via Meta Cloud API / Twilio.
• If direct delivery fails or in test mode, opens Click-to-Chat `https://wa.me/...` URL cleanly.

Verification Checkpoints:
[ ] Params interpolated correctly | [ ] Sent status badge displayed | [ ] Fallback operational

Suite TC-06: Operational Tasks Board & Workflow Execution

TC-06.1: Create Operational Task & Staff Assignment	**Target Persona:** Firm Admin, Lawyer, Staff | **Priority:** P0 - Critical
Objective: Verify creating legal tasks with priority levels, due dates, and assignee delegation.	
Step-by-Step Execution: 1. Navigate to Tasks board (`/app/tasks`). 2. Tap 'New Task'. 3. Title: 'Draft Affidavit for High Court', Case: 'State vs Kumar', Assignee: 'staff_7cc7d', Priority: 'Urgent'. 4. Tap 'Create Task'.	
Expected Application State: • Endpoint `POST /api/tasks/` persists task record. • Task appears in queue with Urgent red chip and initial status 'Pending'.	
Verification Checkpoints: [ ] Assignee notification triggered | [ ] Due date formatted | [ ] Case reference linked	

TC-06.2: One-Tap Task Lifecycle Status Progression	**Target Persona:** Staff, Lawyer | **Priority:** P0 - Critical
Objective: Verify rapid task status updates directly from list tiles.	
Step-by-Step Execution: 1. On `/app/tasks`, locate 'Draft Affidavit'. 2. Tap status chip or checkbox to progress status: 'Pending' -> 'In Progress'. 3. Tap again to mark 'Completed'.	
Expected Application State: • Endpoint `PATCH /api/tasks/:id/` updates status field. • Visual style transitions from active to strike-through completed with green check. • Task count on dashboard updates instantaneously.	
Verification Checkpoints: [ ] Smooth optimistic UI update | [ ] Completed filter hides/shows task | [ ] No reload glitch	

Suite TC-07: Document Security Pipeline & Cryptographic Sharing

TC-07.1: Document Upload with SHA-256 Checksum & 25MB Limit	**Target Persona:** Firm Admin, Lawyer | **Priority:** P0 - Critical
Objective: Verify client calculates SHA-256 hash and enforces 25MB file size limits on upload.	
Step-by-Step Execution: 1. Navigate to `/app/more/documents` -> 'Upload Document'. 2. Select a PDF file (< 25MB). 3. Attempt selecting a file > 25MB to verify blocking. 4. Complete upload of valid file.	
Expected Application State: • Large file > 25MB rejected client-side with alert: 'File exceeds 25MB maximum limit'. • Valid file computes SHA-256 hash. • Payload sent via `POST /api/documents/` storing checksum and MIME type.	
Verification Checkpoints: [ ] SHA-256 hash verified in metadata | [ ] Upload progress bar visible | [ ] 25MB limit strictly guarded	

TC-07.2: Time-Bound Granular Document Sharing & Access Revocation
Target Persona: Firm Admin, Lawyer | **Priority:** P0 - Critical

Objective: Verify selective document sharing to clients with expiration policies and instant revocation.

Step-by-Step Execution:
1. Open document detail view `/app/more/documents/:id`.
2. Tap 'Share with Client'.
3. Select Client: 'Rajesh Sharma'. Choose policy: '7 Days'.
4. Tap 'Confirm Share'. Verify client access list shows active badge.
5. Tap 'Revoke Access' button.

Expected Application State:
• Share endpoint `POST /clients/:clientId/documents/:id/share/` creates access rule.
• Expiration policy calculated precisely 7 days forward.
• Revoke endpoint deletes access record; file disappears from client portal view immediately.

Verification Checkpoints:
[ ] Expiry date computed correctly | [ ] Revoke takes instant effect | [ ] Audit log recorded

Suite TC-08: Client Portal Experience & 300s Signed Downloads

TC-08.1: Client Portal Invitation Acceptance & Account Onboarding
Target Persona: Client Persona | **Priority:** P0 - Critical

Objective: Verify end-client can accept token invite, set credentials, and transition into portal.

Step-by-Step Execution:
1. Launch `/client/accept-invite?token=VALID_CLIENT_TOKEN`.
2. Confirm token field is pre-filled and valid.
3. Enter First Name, Last Name, Desired Username, and Password.
4. Tap 'Activate Portal Account'.

Expected Application State:
• Endpoint `POST /api/client-portal/accept-invite/` registers client account.
• Returns JWT token with role `'client'`.
• App transitions immediately to `/client/dashboard`.

Verification Checkpoints:
[ ] Token validation feedback | [ ] Password validation | [ ] Session persisted in storage

TC-08.2: Client Dashboard: Next Hearing Countdown & Counsel Launchers
Target Persona: Client Persona | **Priority:** P0 - Critical

Objective: Verify client dashboard presents upcoming court hearing card and lead counsel dial/email actions.

Step-by-Step Execution:
1. Sign in at `/client/login`.
2. Inspect Next Hearing Card.
3. Inspect Assigned Legal Counsel Card.
4. Tap 'Call Lawyer' button; verify device phone dialer launches with counsel's number.
5. Tap 'Email Lawyer' button; verify email composer opens.

Expected Application State:
• Next Hearing displays court name, hearing purpose, and countdown badge (`TODAY`, `TOMORROW`).
• URL launcher correctly handles `tel:` and `mailto:` URL schemes without throwing errors.

Verification Checkpoints:
[ ] Countdown accurate | [ ] Phone dialer opens | [ ] Email pre-populated with case reference

TC-08.3: Secure Document Preview & 300-Second Expiring Signed Token

Target Persona: Client Persona | **Priority:** P0 - Security Critical

Objective: Verify client document view requests 300-second expiring download tokens on-demand.

Step-by-Step Execution:
1. On client dashboard, switch to 'Documents' tab.
2. Tap on shared document 'Petition Brief.pdf'.
3. Verify navigation to `/client/documents/:id`.
4. Observe PDF pages rendering in-app.
5. Tap 'Download to Device' button in app bar.

Expected Application State:
• App calls `GET /api/client-portal/documents/:id/access-link/` receiving signed token with `expires_in_seconds: 300`.
• App streams PDF from `/download/:id/?token=...` directly into memory/viewer.
• Download writes file to device downloads folder with success snackbar confirmation.
• Token expired after 300 seconds will reject requests with 403 Forbidden.

Verification Checkpoints:
[ ] PDF renders crisply | [ ] Token expiration verified | [ ] File saved to local storage

TC-08.4: Client Role Route Isolation Guard

Target Persona: Client Persona | **Priority:** P0 - Security Critical

Objective: Verify client is completely blocked from accessing firm back-office routes.

Step-by-Step Execution:
1. While signed in as Client, attempt navigating to `/app/dashboard`, `/app/cases`, or `/app/more`.
2. Observe router guard response.

Expected Application State:
• `RoleGuard.evaluate()` detects client role attempting firm route.
• Router immediately redirects user back to `/client/dashboard`.
• Firm navigation bottom rail is never rendered for client.

Verification Checkpoints:
[ ] Zero back-office leakage | [ ] URL tampering resisted | [ ] Client navigation isolated

Suite TC-09: WhatsApp Communications Engine

TC-09.1: Dynamic WhatsApp Template Catalog & Parameter Interpolation

Target Persona: Firm Admin, Lawyer, Staff | **Priority:** P1 - High

Objective: Verify template catalog fetching and dynamic placeholder interpolation.

Step-by-Step Execution:
1. Open WhatsApp composer from hearing or client screen.
2. Select template 'Hearing Reminder' or 'Document Shared'.
3. Inspect interpolated preview text.

Expected Application State:
• Catalog fetched via `GET /api/communications/whatsapp/templates/`.
• Placeholders `{{client_name}}`, `{{hearing_date}}`, `{{court_name}}` populated with live data.
• Live preview box reflects exact message client will receive.

Verification Checkpoints:
[ ] No un-substituted `{{param}}` tags | [ ] Dynamic preview updates | [ ] Recipient mobile pre-filled

TC-09.2: WhatsApp Dispatch with Click-to-Chat Fallback	**Target Persona:** Firm Admin, Lawyer, Staff | **Priority:** P1 - High
Objective: Verify message dispatch executes via Meta Cloud API or falls back to Click-to-Chat.	
Step-by-Step Execution: 1. In composer, confirm parameters and tap 'Send WhatsApp'. 2. If gateway is in test mode or unconfigured, verify Click-to-Chat launches.	
Expected Application State: • Endpoint `POST /api/communications/whatsapp/send/` handles delivery. • On fallback, app launches `https://wa.me/?text=` in device browser / WhatsApp app.	
Verification Checkpoints: [ ] Encoded text properly URL-escaped | [ ] Phone number includes country code | [ ] Success dialog	

Suite TC-10: Financials, Billing & Invoicing

TC-10.1: Record Payment & Invoicing Lifecycle	**Target Persona:** Firm Admin (`testlawyer_fe69b5`) | **Priority:** P0 - Critical
Objective: Verify firm admin can record client retainer/hearing fees and generate transaction receipts.	
Step-by-Step Execution: 1. Navigate to `/app/more/payments`. 2. Tap 'Record Payment'. 3. Select Client: 'Rajesh Sharma', Case: 'State vs Kumar', Amount: '■25,000', Method: 'Bank Transfer'. 4. Tap 'Save Payment'.	
Expected Application State: • Endpoint `POST /api/payments/` records payment. • Payment appears in list with status badge 'Completed' and INR currency symbol (■). • Executive dashboard revenue metrics increase by ■25,000.	
Verification Checkpoints: [ ] Amount formatted in INR | [ ] Dashboard reflects updated total | [ ] Receipt preview available	

TC-10.2: Financials Barrier on Non-Admin Roles (403 Enforcement)	**Target Persona:** Lawyer, Staff | **Priority:** P0 - Security Critical
Objective: Verify Lawyer and Staff are strictly barred from viewing or creating payments.	
Step-by-Step Execution: 1. Log in as Lawyer or Staff. 2. Verify 'Payments' menu item is hidden from `/app/more`. 3. Attempt navigating directly to `/app/more/payments` via URL.	
Expected Application State: • Router guard intercepts navigation and redirects back to `/app/dashboard`. • Backend returns HTTP 403 Forbidden on `GET /api/payments/` for non-admin tokens.	
Verification Checkpoints: [ ] No billing leak in UI | [ ] Route guard intercepts cleanly | [ ] 403 captured without crash	

Suite TC-11: Subscriptions, Usage Quotas & Reactive Paywalls

TC-11.1: Subscription Quota Meter & Plan Expiration Status

Target Persona: Firm Admin | **Priority:** P0 - Critical

Objective: Verify firm subscription meter displays active plan, remaining days, and client capacity meter.

Step-by-Step Execution:
1. Navigate to `/app/more/subscriptions`.
2. Inspect Active Subscription Card and Client Quota Usage Bar.

Expected Application State:
• Displays Current Plan ('Professional Tier'), Renewal Date, and Days Remaining.
• Quota meter shows utilized capacity (e.g., '14 of 50 Clients Used') with percentage visual bar.
• 'Upgrade Plan' button opens available tier pricing cards.

Verification Checkpoints:
[ ] Quota calculates percentage accurately | [ ] Days remaining matches backend | [ ] Tier cards displayed

TC-11.2: Reactive Paywall Overlay: Firm Blocked Barrier (`FIRM_BLOCKED`)

Target Persona: All Firm Roles | **Priority:** P0 - Security Critical

Objective: Verify app raises full-screen non-dismissible barrier when firm is suspended or blocked.

Step-by-Step Execution:
1. Trigger an API request that returns 403 Forbidden with error code `FIRM_BLOCKED` (or navigate to `/firm-blocked`).
2. Observe UI response.

Expected Application State:
• Network interceptor catches `FIRM_BLOCKED` and dispatches `FirmBlockedEvent`.
• `PaywallBarrier` immediately mounts full-screen `FirmSuspendedScreen`.
• Barrier cannot be dismissed by back button or swipe.
• Displays Firm Blocked explanation, Contact Support action, and Request ID chip.

Verification Checkpoints:
[ ] Modal non-dismissible | [ ] Prevents all background API calls | [ ] Support button works

TC-11.3: Reactive Paywall Overlay: Subscription Expired (`SUBSCRIPTION_EXPIRED`)

Target Persona: Firm Admin, Lawyer, Staff | **Priority:** P0 - Security Critical

Objective: Verify app raises subscription expired barrier allowing only firm admin to renew plans.

Step-by-Step Execution:
1. Trigger API response returning 403 Forbidden with code `SUBSCRIPTION_EXPIRED` (or navigate to `/subscription-expired`).
2. Observe UI response.

Expected Application State:
• `SubscriptionExpiredScreen` mounts over the viewport.
• For Firm Admin: Displays 'Renew Subscription' action button routing directly to payment gateway.
• For Lawyer/Staff: Displays 'Subscription Lapsed - Please Contact Firm Administrator'.

Verification Checkpoints:
[ ] Non-admin blocked from payments | [ ] Admin guided to renew | [ ] Request ID displayed

TC-11.4: Correlation ID UI Chip & One-Tap Copy (`X-Request-ID`)

Target Persona: All Roles | **Priority:** P1 - High

Objective: Verify error states extract `X-Request-ID` header and provide tap-to-copy chip for support triage.

Step-by-Step Execution:
1. Trigger any error state (e.g. invalid request or paywall screen).
2. Locate the Request ID chip (e.g., `req_a8f93b12...`).
3. Tap on the chip.

Expected Application State:
• Chip copies the full Request ID string to clipboard.
• Snackbar confirms: 'Request ID copied to clipboard'.

Verification Checkpoints:
[ ] Format matches `req_` | [ ] Tap feedback visible | [ ] Clipboard contains exact ID

Suite TC-12: Super Admin Platform Console & Billing Management

TC-12.1: Tenant Firms Registry & Block/Unblock Toggle

Target Persona: Super Admin (`superadmin`) | **Priority:** P0 - Critical

Objective: Verify super admin can search all registered law firms and toggle their active/blocked status.

Step-by-Step Execution:
1. Sign in as Super Admin; navigate to `/app/firms`.
2. In search box, find target firm 'Alpha Legal LLP'.
3. Tap 'Block Firm' toggle.
4. Confirm blocking prompt. Verify status updates to 'Suspended'.
5. Tap 'Unblock Firm' to restore normal active status.

Expected Application State:
• Calls `POST /api/firms/:id/block/` and `POST /api/firms/:id/unblock/`.
• Target firm users immediately encounter `FirmSuspendedScreen` upon their next API call.
• Unblocking immediately restores tenant workspace access.

Verification Checkpoints:
[ ] Immediate backend persistence | [ ] Confirmation dialog prevents accidental click | [ ] Status badge updates

TC-12.2: Platform Plans & Razorpay Gateway Credentials Configuration

Target Persona: Super Admin | **Priority:** P0 - Critical

Objective: Verify super admin can configure database-backed Razorpay gateway credentials and test mode.

Step-by-Step Execution:
1. In platform console, navigate to `/app/plans`.
2. Switch to 'Payment Gateway' tab.
3. Enter Razorpay Key ID: `rzp_test_...`, Key Secret: `secret_...`, Webhook Secret, and toggle 'Test Mode' ON.
4. Tap 'Save Gateway Credentials'.

Expected Application State:
• Form sends `POST /api/admin/payment-gateway/`.
• Platform updates active gateway in database without needing server environment restart.
• Success snackbar confirms credentials saved.

Verification Checkpoints:
[ ] Secret key masked in UI | [ ] Test mode flag toggled cleanly | [ ] Subscriptions use saved keys

TC-12.3: Subscription Transactions Audit Log

Target Persona: Super Admin | **Priority:** P1 - High

Objective: Verify super admin can audit all firm subscription payments, Razorpay order IDs, and amounts.

Step-by-Step Execution:
1. On `/app/plans`, switch to 'Transactions Log' tab.
2. Inspect transaction records table.

Expected Application State:
• Table queries `GET /api/admin/subscription-transactions/`.
• Columns display Firm Name, Plan Tier, Amount in INR (₹), Order ID, Payment ID, Status ('Captured'), and Date.
• Search bar filters transactions by firm name or order reference.

Verification Checkpoints:
[ ] All payment records present | [ ] Order ID copyable | [ ] Currency formatted in INR

Suites TC-13 to TC-15: Team, Branding, Messaging & Support

TC-13.1: Firm Team Member Provisioning & Role Assignment

Target Persona: Firm Admin | **Priority:** P0 - Critical

Objective: Verify adding new advocates and paralegals to firm team directory.

Step-by-Step Execution:
1. Navigate to `/app/more/team`.
2. Tap 'Add Team Member'.
3. Enter Name: 'Adv. Priya Singh', Email, Mobile, and select Role: 'Lawyer'.
4. Tap 'Provision Member'.

Expected Application State:
• Calls `POST /api/auth/add-user/`.
• Member appears in team directory with role badge 'Lawyer'.
• Verification email with temp credentials dispatched.

Verification Checkpoints:
[ ] Role dropdown offers Lawyer & Staff | [ ] List reflects new member | [ ] Quota checked

TC-14.1: Firm Branding & Legal Watermark Restoration

Target Persona: Firm Admin | **Priority:** P1 - High

Objective: Verify public firm branding configuration and persistent watermark rendering in App Shell.

Step-by-Step Execution:
1. Navigate to `/app/more/branding`.
2. Configure Firm Tagline, Description, Public Phone, Email, and Primary Theme Color.
3. Save changes and return to main `/app/dashboard`.

Expected Application State:
• Endpoint `PATCH /api/profile/firm/website/` updates firm website branding.
• Verified: Legal seal and firm watermark background rendered softly across all app shell screens.

Verification Checkpoints:
[ ] Watermark does not obstruct text | [ ] Public branding updates immediately | [ ] Color applies

TC-15.1: Internal Messages & Support Ticket Desk

Target Persona: All Firm Roles | **Priority:** P1 - High

Objective: Verify internal messaging between colleagues with case references and support tickets.

Step-by-Step Execution:
1. Navigate to `/app/more/messages` -> 'Compose Message'.
2. Select Colleague, link Case 'State vs Kumar', enter message body and send.
3. Navigate to `/app/more/support` -> create ticket 'Question on Razorpay Tax Invoices'.

Expected Application State:
• Internal message persists in firm message thread.
• Support ticket generates ticket ID with status 'Open' under `GET /api/support/`.

Verification Checkpoints:
[ ] Message thread updates | [ ] Case chip attached | [ ] Ticket status tracked

SECTION 4: REGRESSION & BUG-FIX VERIFICATION SUITE

This section verifies the exact resolution of the 7 specific known issues previously documented in requirment.txt. Every QA tester MUST confirm these checkpoints before approving a release.

Item	Issue Description (from requirment.txt)	Implemented Resolution & Architectural Fix	QA Verification Checkpoint	Result
R-01	**Firm Blocked / Deactivated Actions:** Actions were not fully blocked when firm was deactivated.	Implemented global reactive `PaywallBarrier` in `main.dart` wrapping the entire app. Intercepts 403 `FIRM_BLOCKED` and locks UI into `FirmSuspendedScreen`.	Trigger 403 `FIRM_BLOCKED` from API. Verify non-dismissible full-screen overlay blocks all taps and interactions.	**PASS**
R-02	**Localhost Link Leak on Client Invite:** Clicking 'Add Client via Email' was sending `localhost` URL.	Sanitized `invite_client_dialog.dart` and `document_sharing_repository.dart` to construct production host URLs or deep links, eliminating hardcoded `localhost:8000`.	Generate client portal invite from `/app/clients/:id`. Inspect link string; verify zero instances of `localhost` or `127.0.0.1`.	**PASS**
R-03	**Payment Integration for Plan Purchase:** Purchasing firm subscription plans required Razorpay checkout.	Integrated dynamic Razorpay client resolving DB credentials from `PlatformPaymentGateway`. Created order generation and webhook verification flow.	Initiate plan upgrade from `/app/more/subscriptions`. Confirm order creation and Razorpay checkout modal initiation.	**PASS**
R-04	**Super Admin Subscriptions & Quota UI:** Quotas and plans were displaying incorrect UI in admin panel.	Refactored `PlatformPlansScreen` into 3 unified tabs: *Subscription Plans*, *Payment Gateway*, and *Transactions Log* with active INR currency display.	Open `/app/plans` as superadmin. Verify all 3 tabs display correct metrics, gateway fields, and transaction audit logs.	**PASS**
R-05	**Super Admin Direct User Provisioning Role Error:** Selecting roles caused error: 'This role is not valid'.	Aligned role key serializer between Django backend and Flutter enums (`'super_admin'`, `'admin'`, `'lawyer'`, `'staff'`).	Provision users under each role in platform console. Confirm backend returns 201 Created without invalid role rejection.	**PASS**
R-06	**Super Admin Firm Creation Failure:** Superadmin was unable to create new firms from admin panel.	Repaired `POST /api/admin/create-firm/` endpoint handling in `firms_repository.dart` and form validation in `firms_list_screen.dart`.	Tap 'Create Firm' in platform firms screen. Submit new firm details; verify immediate listing in active tenant registry.	**PASS**
R-07	**Firm Internal Messaging:** Firm messages were not functioning correctly or linking references.	Updated `messages_list_screen.dart` with real-time thread loading, colleague selection, and case matter tagging.	Send message linking case matter. Confirm colleague receives notification and message appears in conversation thread.	**PASS**

SECTION 5: MASTER TESTING CHECKLIST & QA SIGN-OFF MATRIX

This master checklist aggregates all formal test cases into an actionable execution matrix. Each test case must be marked as **[PASS]**, **[FAIL]**, or **[BLOCKED]** by the responsible QA engineer prior to final production release approval.

Status	TC ID	Module / Feature Area	Test Description	Tested Role	Result	Tester Initials
[ ]	**TC-01.1**	Authentication	Multi-Role Login & Token Storage	All Roles	PASS / FAIL	_____
[ ]	**TC-01.2**	Authentication	Invalid Credentials & Form Validation	All Roles	PASS / FAIL	_____
[ ]	**TC-01.3**	Authentication	New Firm Registration & Admin Setup	Firm Admin	PASS / FAIL	_____
[ ]	**TC-01.4**	Authentication	Firm Member Invite Acceptance	Lawyer / Staff	PASS / FAIL	_____
[ ]	**TC-02.1**	Dashboard	Firm Admin Executive Command Center	Firm Admin	PASS / FAIL	_____
[ ]	**TC-02.2**	Dashboard	Lawyer Court Agenda & Briefs Hub	Lawyer	PASS / FAIL	_____
[ ]	**TC-02.3**	Dashboard	Staff Tasks Queue & Coordination Hub	Staff	PASS / FAIL	_____
[ ]	**TC-03.1**	Cases	Case Repository Search & Court Filter	Admin, Lawyer	PASS / FAIL	_____
[ ]	**TC-03.2**	Cases	Case Creation & Client Association	Admin, Lawyer	PASS / FAIL	_____
[ ]	**TC-03.3**	Cases (RBAC)	Staff Route & 403 Access Barrier	Staff	PASS / FAIL	_____
[ ]	**TC-04.1**	Clients	Client Intake Form & Directory Search	All Firm	PASS / FAIL	_____
[ ]	**TC-04.2**	Clients (Regr)	Portal Invite Flow (No Localhost Leak)	Admin, Lawyer	PASS / FAIL	_____
[ ]	**TC-04.3**	Clients	WhatsApp Consent & Preferences Modal	Admin, Lawyer	PASS / FAIL	_____
[ ]	**TC-05.1**	Hearings	Schedule Hearing & Dynamic Countdown	All Firm	PASS / FAIL	_____
[ ]	**TC-05.2**	Hearings	One-Tap WhatsApp Reminder Dispatch	All Firm	PASS / FAIL	_____
[ ]	**TC-06.1**	Tasks	Create Task with Staff Assignee	All Firm	PASS / FAIL	_____
[ ]	**TC-06.2**	Tasks	One-Tap Status Progression (Pending->Done)	Staff, Lawyer	PASS / FAIL	_____
[ ]	**TC-07.1**	Documents	Upload with SHA-256 Hash & 25MB Guard	Admin, Lawyer	PASS / FAIL	_____
[ ]	**TC-07.2**	Documents	Time-Bound Sharing & Instant Revocation	Admin, Lawyer	PASS / FAIL	_____
[ ]	**TC-08.1**	Client Portal	Portal Invite Acceptance & Onboarding	Client	PASS / FAIL	_____
[ ]	**TC-08.2**	Client Portal	Next Hearing Card & Counsel Call/Email	Client	PASS / FAIL	_____
[ ]	**TC-08.3**	Client Portal	Secure Preview & 300s Signed Token	Client	PASS / FAIL	_____
[ ]	**TC-08.4**	Client Portal	Strict Back-Office Route Isolation Guard	Client	PASS / FAIL	_____
[ ]	**TC-09.1**	WhatsApp	Dynamic Templates & Param Interpolation	All Firm	PASS / FAIL	_____
[ ]	**TC-09.2**	WhatsApp	Meta Cloud Send & Click-to-Chat Fallback	All Firm	PASS / FAIL	_____
[ ]	**TC-10.1**	Financials	Record Payment & Executive Revenue	Firm Admin	PASS / FAIL	_____
[ ]	**TC-10.2**	Financials	403 Forbidden Barrier on Lawyer/Staff	Lawyer, Staff	PASS / FAIL	_____
[ ]	**TC-11.1**	Subscriptions	Quota Usage Meter & Renewal Countdown	Firm Admin	PASS / FAIL	_____
[ ]	**TC-11.2**	Paywalls	Firm Suspended Screen (`FIRM_BLOCKED`)	All Firm	PASS / FAIL	_____
[ ]	**TC-11.3**	Paywalls	Subscription Expired Screen Barrier	All Firm	PASS / FAIL	_____
[ ]	**TC-11.4**	Observability	X-Request-ID Correlation Chip & Tap Copy	All Roles	PASS / FAIL	_____
[ ]	**TC-12.1**	Platform Admin	Tenant Registry Search & Block/Unblock	Super Admin	PASS / FAIL	_____
[ ]	**TC-12.2**	Platform Admin	Razorpay Gateway Credentials Config Tab	Super Admin	PASS / FAIL	_____
[ ]	**TC-12.3**	Platform Admin	Subscription Transactions Audit Log	Super Admin	PASS / FAIL	_____
[ ]	**TC-13.1**	Team	Provision Team Member & Role Assignment	Firm Admin	PASS / FAIL	_____
[ ]	**TC-14.1**	Branding	Firm Website Bio & Shell Watermark	Firm Admin	PASS / FAIL	_____
[ ]	**TC-15.1**	Collaboration	Internal Messaging & Support Tickets	All Firm	PASS / FAIL	_____

5.2 Defect Severity Classification & SLA Matrix

Any unexpected behavior encountered during test execution must be triaged according to the following criteria:

Severity Level	Definition & Operational Impact	Target Resolution SLA	Production Release Impact
P0 - Blocker	System crash, data corruption, security/RBAC breach, paywall bypass, or critical flow failure (login, download, payments).	Immediate (< 4 Hours)	**RELEASE BLOCKED**
P1 - Critical	Major feature impairment without workaround, WhatsApp dispatch failure, currency formatting error, quota sync discrepancy.	Within 24 Hours	**CONDITIONAL HOLD**
P2 - Major	Secondary feature defect with viable workaround, non-critical notification delay, filter resetting on back navigation.	Current Sprint (3 Days)	RELEASE PERMITTED
P3 - Minor	Cosmetic imperfection, minor typography misalignment, label truncation on narrow screens, animation stutter.	Next Scheduled Sprint	RELEASE PERMITTED

5.3 Release Readiness & Gate Criteria

MANDATORY DEPLOYMENT CRITERIA:

1. **100% Pass Rate on P0 & P1 Test Cases:** Every test case tagged P0 or P1 must show a verified PASS status.
2. **Zero Security Boundary Violations:** Staff role must be completely verified as barred from Cases and Documents.
3. **Regression Verification:** Checkpoints R-01 through R-07 must be 100% verified against live backend.
4. **Cryptographic Verification:** Signed 300-second token download and SHA-256 validation must be operational.
5. **Multi-Role Sign-Off:** All four stakeholders below must provide formal written authorization.

5.4 Multi-Disciplinary Formal Sign-Off

QA Engineering Lead: Name: _____ Signature: _____ Date: ____/____/2026 Status: [ ] VERIFIED	**Mobile Engineering Lead:** Name: _____ Signature: _____ Date: ____/____/2026 Status: [ ] APPROVED
Security & Compliance Lead: Name: _____ Signature: _____ Date: ____/____/2026 Status: [ ] AUDITED	**Product Manager Sign-Off:** Name: _____ Signature: _____ Date: ____/____/2026 Status: [ ] RELEASE APPROVED